Bulk Density (30.48 cm) Sampling
Supporting Document `IndigoCarbon_US-1_2023_0002`

June 25, 2024

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1 Context

When soil sampling for Indigo Carbon, one fifth (samples collected October 1, 2019 – March 15, 2020) or one third (samples collected March 16, 2020 – present) of carbon sampling locations are designated as Bulk Density sampling locations (see `IndigoCarbon_US-1_2023_0004` for an explanation of the sample point prescriptions).

At each Bulk Density site, three different sample types are collected (Figure 1) and the sample type bolded below must be collected according to the instructions in this document.

- An individual 30 cm carbon sample according to the instructions in `IndigoCarbon_US-1_2023_0001`
- **An individual 30.48 cm bulk density sample according to the work instructions below**
- A core that will contribute to a composite sample for pH and texture testing according to the work instructions in `IndigoCarbon_US-1_2023_0003` (applicable only to sites sampled March 16, 2020 – present)

2 Inventory Checklist

	Tools	Quantity	Vendor	Catalog No.	Required/ Recommended
Bulk Density Sample Collection	2" X 12" SST SCS Cup & Cap (30.48 cm)	2-3	AMS	404.75	**Required** (part and brand)
	2"x12" SST liner	2	AMS	405.14	**Required** (part and brand)
	Slide Hammer	2	AMS	400.99	Recommended
	24" Pipe Extension	1	AMS	408.01	Recommended
	Rubber Coated Cross Handle	1	AMS	406.04	Recommended
	Rags	1 box	Uline	S-7287	Recommended
	Spatula	1	N/A	N/A	Recommended
	Serrated Knife	1	N/A	N/A	Recommended
	Brushes	1	N/A	N/A	Recommended
	Vice Grip	1	N/A	N/A	Recommended
	Large Crescent Wrench	2	N/A	N/A	Recommended
Demolition Hammer Kit	Bulk Density Adaptor	1	Indigo Ag	N/A	Recommended
	Predator 2000 Watt Super Quiet Inverter Generator	1	Harbor Freight	62523	Recommended
	Bauer 12.5 Amp SDS Max Type Pro Demo Hammer Kit	1	Harbor Freight	63437	Recommended
	McGraw 8gal 150 PSI Pro Air Compressor	1	Harbor Freight	64294	Recommended
	CP Air Compressor Starter Kit, 20 Pc	1	Harbor Freight	64599	Recommended
	Vanguard 25ft x 12ga Outdoor Extension Cord	1	Harbor Freight	62948	Recommended
Retrieving Bulk Density Core	Farm Jack	1	Bulldog	7033400	Recommended
	Chain	1	N/A	N/A	Recommended
	Pad	1	N/A	N/A	Recommended
Sample Bagging / Labeling	QR code labels	Roll	Indigo	N/A	**Required**
	Whirl-Pak bag (55 oz) - yellow	1 box (500)	Nasco	B01195	Recommended

Table 1: Checklist of tools and supplies used in bulk density sampling

3 Preparation

- Familiarize yourself with Indigo's field Safety Protocol.

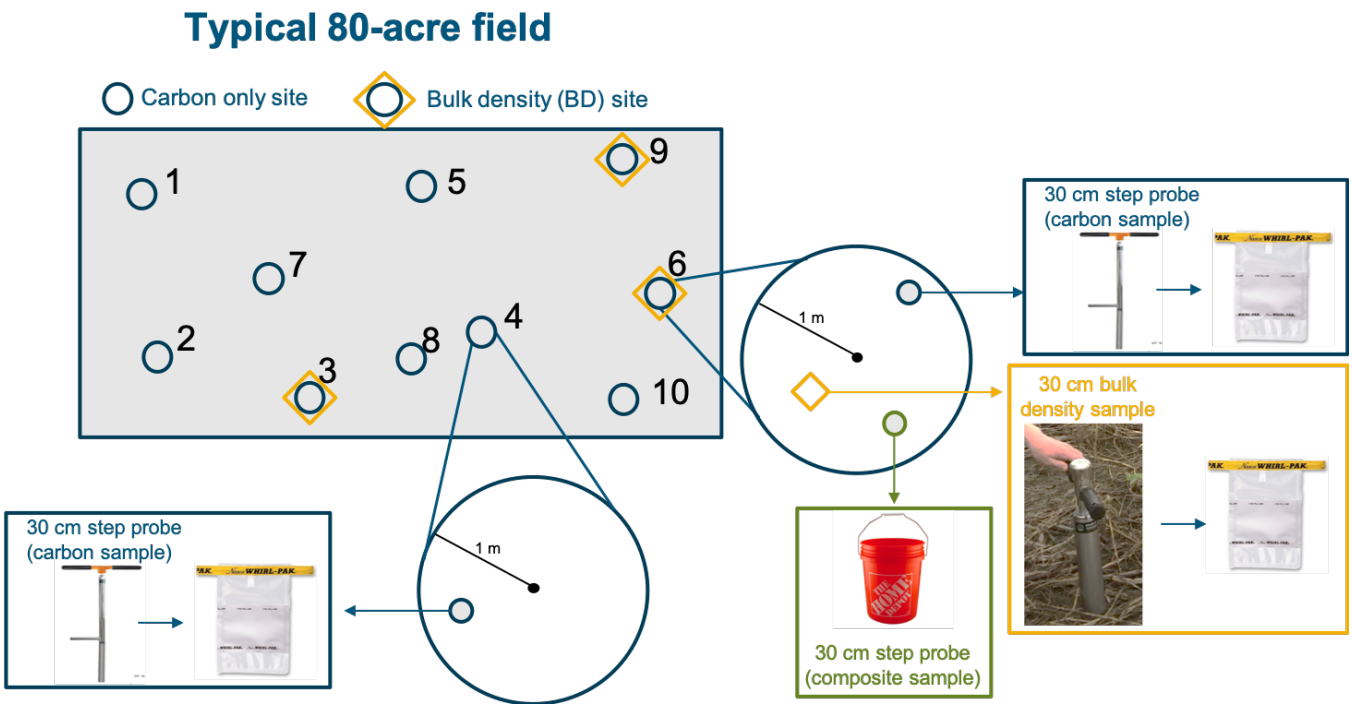

Figure 1: Indigo Carbon soil sample collection

- NOTE: Safety rules and regulations of sampler's parent company should always supersede, when applicable, those recommended by Indigo.
- For this procedure you will need.
  - Safety Glasses
  - Steel Toe Shoes
  - Cut-Resistant Gloves
- Prepare sampling equipment ahead of time and ensure that you have all of the necessary equipment included in the checklist above.
  - Take note of supplies in short inventory and request additional consumables from your Equipment Manager.
- Carbon sampling locations designated as Bulk Density sampling locations are explicitly labeled as such within Indigo Fields 2.0

4 Sampling

Navigation and site preparation

- Navigate to the appropriate sampling location by following the instructions outlined in `IndigoCarbon_US-1_2023_0004`.
  - There are specific instructions in `IndigoCarbon_US-1_2023_0004` that will guide you when faced with situations when you are unable to navigate to the prescribed point due to any reason (flood, fence, private property) or find that the point is in a compromised sampling location (field edge, access road, rock outcropping).
- Clear residue or vegetation from the soil surface where you intend to take a soil core. Only sample bare mineral soil and avoid residue in the sample (Figure 2).

Assessment of Soil Conditions

Figure 2: Residue clearing to prepare sample core extraction

- Excessively wet soils
  - Do not sample in standing water of any depth
  - If upon removal of a soil core, the hole created is refilled within two minutes by surrounding soil flowing back into the hole because the soil exhibits a fluid-like property, discard the sample and discontinue sampling
- Frozen soils
  - To avoid erroneous measurements of bulk density, we do not want to sample frozen soils
  - When the daily average air temperature falls below 32°F, consult the instructions at the bottom of this page to determine if samples can be taken

4.1 Core Sampling

NOTE: The core method is the primary method of sampling, only proceed with the alternative method once the core method has been tried and proven unsuccessful multiple times.

Prepare the sampling equipment

- Mark the outside of the bulk density sampling device with tape or a marker to indicate the position of the top of the internal liner
- **Make sure there is no soil or debris on the walls of the cup or in the threads**. If there is, remove the soil with a wire brush, pipe brush, or cloth.
  - Soil/debris within the threads will seize up the cap on the bulk density core and make unscrewing it extremely difficult after pulling the core up from the ground.
  - Soil/debris within the cup will not only contaminate the sample, but will make removing the metal liner from the corer more difficult
- Place the 30.48 cm stainless steel liner inside the cup.
- Screw on cap

Slide hammer driving method. Prepare to drive corer into the ground.

- Screw slide hammer to top of cap.
  - **The slide hammer does not lock – be careful that it does not slide unexpectedly and pinch your skin.**
- Place BD sampler in a rock-free spot of ground on bare mineral soil.
- Keep feet shoulder-width apart with a slight bend in the knees.

- Lift the slide hammer up and bring it down with your weight while ensuring the bulk density corer is being driven straight into the ground and not at an angle.
- Proceed to "Completing the bulk density sample".

Demolition hammer driving method

Figure 3: Generator starting procedure.

- Turn the yellow knob on top of the generator to "ON" (figure 3).
- Turn the dial on the front panel of the generator to "Start".
- Firmly pull the rip cord on the side of the generator.
- Once generator has been running for several seconds, switch the dial on the front panel to "Run" (figure 3).
- Make sure the bulk density adaptor is attached to the driving head of the demolition hammer.
- Flip the power switch on the demolition hammer to "ON".
- Place BD sampler in a rock-free spot of ground on bare mineral soil. Push or drive it far enough into the ground that it can stand on its own (Figure 4a).
- Fit the bulk density adaptor on the demolition hammer over top of the bulk density cap (Figure 4b).
- Pull the trigger to initiate a hammering action that will drive the sampler into the ground, keeping your feet shoulder-width apart with a slight bend in the knees.
- Proceed to "Completing the bulk density sample"

Completing the bulk density sample

- Continue to drive the corer into the ground **until you reach the line marked on the top of the corer cylinder indicating the top of the corer liner (~1 cm above the soil surface)** to avoid compacting soil against the top of the cap (Figure 5)
- The soil surface should be flush with the top of the liner across the entirety of the cross-sectional area. To ensure this result, it is acceptable to sample slightly above the liner rim (until the lowest point of the soil surface reaches the rim), and scrape off the excess portions beyond the rim as long as the **soil surface does not reach the bottom of the cap.**

Figure 4: Bulk density sample extraction with corer. From left to right: a, b, c

Figure 5: Bulk density corer driven into ground at proper depth

- If you notice that the corer has stopped moving into the soil, it is likely that it has hit a rock. If this is the case, you will need to extract the corer and reattempt sample collection within 12 inches (30.48 cm) of the original attempt two more times. If those attempts are unsuccessful, relocate according to "Obstacles and Alternate Locations" in `IndigoCarbon_US-1_2023_0004`.
  - **Note: Rocks tend to cause more vibration in the handle of the slide hammer than hardpans. Ensure the obstacle is a rock before selecting an alternate location.**
  - When removing the corer, make sure to fully clean it out before starting again

Extract the corer from the ground

- **Bulldog Farm Jack Device** (*Review jack manual before operating)
  - Remove the core driving device from the corer.
  - Replace the hammer head with the cross handle.
  - Grab the cross handle and rotate the corer in the ground to release any suction of the soil on the corer. Rock corer back and forth to separate core from soil at the bottom of the excavated hole.
  - Retrieve jack from UTV with chain and floor plate.
  - Place floor plate within 3" of bulk density corer.
  - Place the jack on the floor plate where the tongue is facing the bulk density corer (figure 6a).
  - Wrap the chain under both handles of the Hammer Head Cross.
  - Connect the chain around the tongue of the jack, eliminating as much slack as possible (Figure 6b).
  - Use the hand lever to jack the bulk density corer from the soil (Figure 6c).
  - Once the corer is halfway out of the ground, physically remove the rest of the corer by hand.
  - Safely return jack and floor plate to UTV.

Figure 6: Extraction of bulk density core with jack. From left to right: a, b, c

Excavating the bulk density sampler from the ground

- In some instances, using the Bulldog Jack to remove the sample may be problematic.
  - In sandy soils, the core may not hold together, and soil particles may leak out the bottom of the corer.
  - In wet and clayey soils, there may be enough surface tension between the sample within the corer and the soil still in the ground that the sample is sucked out of the corer during the extraction.
- In such cases, the corer must be excavated from the ground.
  - Dig around the corer to a depth of approximately 6-8 inches on all sides (Figure 7).
  - When extracting, tilt the corer at a slight angle to prevent the sample from sliding out.

Figure 7: Excavating problematic bulk density samples

Extract the sample from the corer

- Lay corer flat on the ground.
- Unscrew the cap of the bulk density corer.
  - If the cap is tight:
    - Tap the upper portion of the corer where the threads are located to loosen the threads.
    - Utilize the wrench where the extension pipe and the cap threads meet to ensure they will not move and twist the corer counterclockwise (not applicable for samples pulled with the Bulldog Jack).
    - Place the wrench on the cap and lightly tap the wrench clockwise to loosen the threads (Figure 8).

Figure 8: Removing a tight bulk density cap

- Place a hard, flat surface at the sampling liner (i.e., the end where you just removed the cap). This can be a paint scraper or a blue plastic cap.
- Turn the corer upside down and pull the bulk density corer off of the liner.
  - You may need to knock lightly on the metal rim of the corer or press gently on the soil at the bottom of the corer to dislodge the liner from the sampler.
  - Place the end of the liner covered by a flat surface on the ground.

- Steady the liner with one hand while using a serrated/putty knife to cut/scrape off any additional sample that extends beyond the liner to create a flush surface (Figure 9). **Again, be careful to avoid compacting the sample!**.

Figure 9: Scraping of additional sample to create a flush surface

- Confirm that there are no irregularities, large rocks, or dead spaces on that end of the core.
- Bag the sample.
  - Place the end of the Whirl-Pak over the top of the exposed end of the core, invert, and remove the blue plastic cap/hard surface.

Figure 10: Bagging the bulk density sample

- Use a putty knife to ensure that the soil is flush with the liner on this end of the cap as well.
- Push sample into the bag. **Ensure that the entire sample is transferred to the bag - accurate volume is critical.**
  - It does not matter if the core remains intact once it has left the liner, the crucial part is that the liner was filled with soil, flush on both ends with the liner rim, and that the full volume of soil from the liner was transferred to the bag.

- Clean up all equipment before next use.
  - The sampling liner, plastic cap, metal cap, spatula and knife should be brushed and the inside wiped with a rag, particularly near the threads.
  - If you needed to dig out the corer, try to replace as much soil into the hole as possible to minimize field disturbance.

5 Sample Labeling

- Make sure there is a QR code affixed to the bag.
  - Note: if bag is wet, wipe off before attempting to add sticker, otherwise it may fall off during shipping.
- With a sharpie write the following information as it appears in Indigo Fields 2.0 (IF 2.0) on the white section of the Whirl-Pak on the back of the bag (opposite from QR code label):

Minimum recommended labeling system:

- Sample Date (month/day)
- Field Name "Unique Full Word" (3 character minimum unless entire name is less).
- No multi-word abbreviations.
  - From the following example:
    - For the field: Craig Place 1 - use "Place 1" **not** CP1
    - For the field: K Pivot SC - use "Pivot SC" **not** KPSC
- Sample Type (C, BD, pH/TXT) – **one code per bag**
- Sample Number
  - For carbon: C – 1,2,3,4...
  - For bulk density: BD – 1,2,3,4...
- **Include more info. if it helps you (e.g. grower name, farm name, full field name, etc.)**
  - **Minimum requirement example:**
    - For the first bulk density sample collected on the field Crabbe C6C on 5/13:
      - 5/13 - C6C - BD1
  - **More info. example:**
    - For a carbon sample collected at the fifth spot label as follows. **Grower**: Fred **Farm**: McDonalds **Field**: Home **Sample:** C5
      - Fred-McDonalds-Home-C5

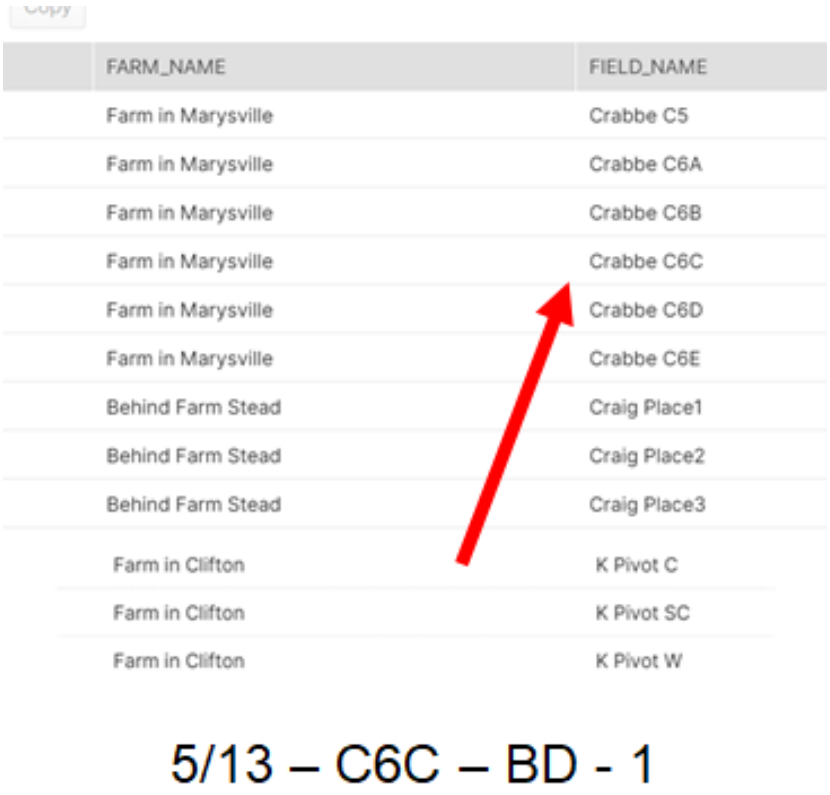

Figure 10. Labeling the bulk density sample

6 Record the sample in IF 2.0

- The Carbon sample for this sampling location should already have been recorded in IF 2.0 according to the work instructions in `IndigoCarbon_US-1_2023_0001`
- For point where a bulk density sample is required in addition to an SOC sample, there will be additional survey questions after the Carbon questions (figure 11).
  - Enter information for the general point as described in Soil Carbon (30 cm) Sampling.
  - Enter the Carbon core information (QR code, related data).
  - Enter the bulk density core information.
    - If you are unable to take a core, select a valid reason why not. We ask whether you were able to take each type of sample - so if you were able to take the Carbon core, but not the bulk density core, you could answer each individually.
    - If you are able to take a core, proceed to scan the QR Code (figure 11).
  - For each type of sample we ask that you enter whether you reached the sample depth if you did not reach the full 30 cm, and to enter a reason why the depth was not reached

7 Troubleshooting bulk density sample collection

What to do when you do not collect a full bulk density core

- In particularly sandy, clayey or rocky soils, collecting a fully intact 30.48 cm bulk density core can be challenging; however, this measurement is crucial for our calculation of carbon stocks
- If on your first attempt to collect a bulk density sample you are unsuccessful:

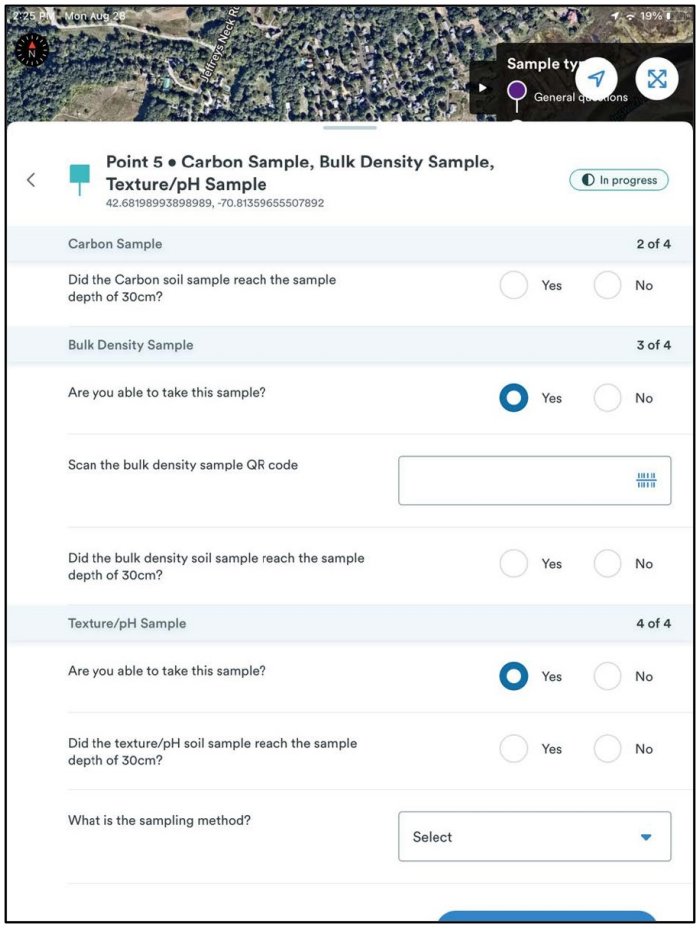

Figure 11: Indigo Fields 2.0 form for QR code and sample

- Measure the length of the collected sample within the core liner and make a note of the length on the bag
- Bag the sample and set aside
- Attempt to collect a bulk density sample to full depth **two additional times** at the prescribed location
  - NOTE: Always save the deepest sample obtained until a deeper or full depth sample is collected
- If a full depth sample cannot be collected at the prescribed location after **three attempts** (offsetting bullet point only applies to samples collected March 1, 2021 – present):
  - Follow the offsetting protocol listed in the "**Obstacles and Alternate Locations"** section `IndigoCarbon_US-1_2023_0004` to attempt to locate a sampling point in which a full depth core can be extracted. One attempt should be made at each new offset location to obtain a sample at full depth sampling is completed.
    - **NOTE: Carbon samples, bulk density, and pH/texture samples must be collected at the same location. If the bulk density sample is offset, the carbon and pH/texture composite samples must be obtained from the new offset location as well.**
- If a full depth sample cannot be achieved:
  - Mark the QR code, sampling location, and depth of the deepest sample collected in the appropriate sections of the IF "Soil Sampling" form and proceed to the next prescribed sampling location.

8 Summary of Changes

Section	Date	Summary of Changes
Throughout	May 12, 2024	Changes made to references made to legacy version if Indigo Fields Mobile